Sekwon Lee

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Research Interest

Computer systems: Storage/Memory systems, Distributed systems, Operating systems, Database systems

Focus: Next-generation systems for emerging memory/storage and disaggregation (RDMA, CXL) technologies

- Indexing, Caching, Tiering, Concurrency, Crash consistency, Fault tolerance

Education

University of Texas at Austin — *Austin, TX, US*
Ph.D. in Computer Science — *08/16/2018 - 12/16/2023*

- Advisor: Vijay Chidambaram
- Dissertation: Designing Key-Value Stores for Emerging Memory and Disaggregation Technologies

UNIST (Ulsan National Institute of Science and Technology) — *Ulsan, South Korea*
M.S. in Computer Science and Engineering — *03/01/2016 - 02/12/2018*

- Advisor: Sam H. Noh
- Thesis: Write-Optimal Radix Tree: A Deterministic Indexing Structure for Persistent Memory Storage Systems

Hongik University — *Seoul, South Korea*
B.S. in Computer Engineering — *03/01/2009 - 02/23/2015*

- Undergraduate advisor: Sam H. Noh

Work Experience

HPE Labs — *Seattle, WA, US (Remote)*
Research Engineer — *01/08/2024 - present*

- Job description: Research, design, and implement a distributed tiered caching system. Lead the design and development of the far-memory tier, including concurrency control mechanisms, cache eviction algorithms, and cache sharing and tiering policies. Lead research to adapt the system for KV cache offloading in LLM inference stacks and evaluate the benefits of this approach.

Microsoft Research — *Austin, TX, US (Remote)*
Research Intern — *05/24/2021 - 08/13/2021*

- Job description: Worked on scale-out AMBROSIA, a general framework to build resilient distributed systems. Implemented sharding supports with functions to filter out RPC requests and log entries irrelevant to the corresponding shard membership.
- Mentor: Jonathan Goldstein

Hewlett Packard Labs — *Palo Alto, CA, US*
Research Associate Intern — *06/03/2019 - 08/23/2019*

- Job description: Designed far-memory data structures optimized for one-sided RDMA operations. Designed and implemented a hybrid index structure combining a prefix trie with hash tables to take both advantages of an easily cacheable trie structure and one-sided RDMA-efficient hash tables.
- Mentors: Kimberly Keeton, Sharad Singhal, and Marcos K. Aguilera

Hewlett Packard Labs *Palo Alto, CA, US*

RESEARCH ASSOCIATE INTERN *06/26/2017 - 09/22/2017*

- Job description: Designed a DRAM cache for key-value stores working on FAM (Fabric-Attached Memory). Designed and implemented a hybrid approach that caches both value and shortcut entries.
- Mentors: Kimberly Keeton, Haris Volos, and Yupu Zhang

Publications

17 peer-reviewed publications: 9 conference papers (SOSP, VLDB, FAST, MICRO, etc.) and 8 peer-reviewed workshop papers, including a memorable paper award at NVMW.

Conferences

[1] Clarete Riana Crasta, John L Byrne, Abhishek Dwaraki, David Emberson, Harumi Anne Kuno, **Sekwon Lee**, Ramya Ahobala Rao, Shreyas Vinayaka Basri K S, Amitha Chandrachari, Chinmay Ghosh, Rishi Kesh K Rajak, Sriram Ravishankar, Porno Shome, and Lance Evans, **Global Distributed Client-side Cache for DAOS**, Proceedings of the 2025 Cray User Group (CUG 2025).

[2] **Sekwon Lee**, Soujanya Ponnapalli, Sharad Singhal, Marcos K. Aguilera, Kimberly Keeton, and Vijay Chidambaram, **DINOMO: An Elastic, Scalable, High-Performance Key-Value Store for Disaggregated Persistent Memory**, Proceedings of the VLDB Endowment, Volume 15, Issue 13 (VLDB 2023).

[3] **Se Kwon Lee**, Jayashree Mohan, Sanidhya Kashyap, Taesoo Kim, and Vijay Chidambaram, **RECIPE: Converting Concurrent DRAM Indexes to Persistent-Memory Indexes**, Proceedings of the 27th ACM Symposium on Operating Systems Principles (SOSP 2019).

[4] Rohan Kadekodi, **Se Kwon Lee**, Sanidhya Kashyap, Taesoo Kim, Aasheesh Kolli and Vijay Chidambaram, **SplitFS: Reducing Software Overhead in File Systems for Persistent Memory**, Proceedings of the 27th ACM Symposium on Operating Systems Principles (SOSP 2019).

[5] Qingrui Liu, Joseph Izraelevitz, **Se Kwon Lee**, Michael L. Scott, Sam H. Noh, and Changhee Jung, **iDO: Compiler-Directed Failure Atomicity for Nonvolatile Memory**, Proceedings of the 51st Annual IEEE/ACM International Symposium on Microarchitecture (MICRO 2018).

[6] **Se Kwon Lee**, K. Hyun Lim, Hyunsub Song, Beomseok Nam, and Sam H. Noh, **WORT: Write Optimal Radix Tree for Persistent Memory Storage Systems**, Proceedings of the 15th USENIX Conference on File and Storage Technology (FAST 2017).

[7] Hyunsub Song, Young Je Moon, **Se Kwon Lee** and Sam H. Noh, **PMAL: Enabling Lightweight Adaptation of Legacy File Systems on Persistent Memory Systems**, Proceedings of the 2017 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2017).

[8] **Se Kwon Lee**, Hyunsub Song, Young Je Moon and Sam H. Noh, **Experimental Evaluation of File System Data Structures for New Memory based Storage**, Proceedings of the 2016 Korea Computer Congress (KCC 2016, **Best Paper Award**).

[9] Hyunsub Song, Young Je Moon, **Se Kwon Lee** and Sam H. Noh, **Lightweight Adaptation of Legacy File Systems for Persistent Memory based Storage**, Proceedings of the 2016 Korea Computer Congress (KCC 2016, **Best Paper Award**).

Workshops

[1] Abhishek Dwaraki, Sreenivas R. Sukumar, Christopher D. Rickett, Clarete Riana Crasta, Harumi Kuno, Michael Neal, Pankaj Pandey, Ryan Yates, Karlon West, Amar Gopal Chittiboyina, Ikhlas A. Khan, John Byrne, **Sekwon Lee**, and David Emberson, **A Query Engine for Scientific Data Exploration Using Theory, Simulation, and Artificial Intelligence Models**, Proceedings of the Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops 2025, AI4S).

[2] John Byrne, Clarete Crasta, Abhishek Dwaraki, David Emberson, Harumi Kuno, **Sekwon Lee**, Sharad Singhal, Ramya Ahobala Rao, Shreyas Vinayaka Basri K S, Amitha C, Chinmay Ghosh, Rishi Kesh Rajak, Sriram Ravishankar, Porno Shome, Lance Evans, Sherin George, Kevan Rehm, Myungjun (MJ) Son, Taeklim Kim, Shiyue (Jason) Hou, **A Global In-Memory Cache and Computation Tier for DAOS** (Work In Progress paper), The 9th International Parallel Data Systems Workshop (PDSW 2024).

[3] **Sekwon Lee**, Soujanya Ponnapalli, Sharad Singhal, Marcos K. Aguilera, Kimberly Keeton, and Vijay Chidambaram, **DINOMO: An Elastic, Scalable, High-Performance Key-Value Store for Disaggregated Persistent Memory** (Extended abstract of the VLDB 2023 paper), The 3rd Workshop On Resource Disaggregation and Serverless Computing (WORDS 2022).

[4] **Se Kwon Lee**, Jayashree Mohan, Sanidhya Kashyap, Taesoo Kim, and Vijay Chidambaram, **RECIPE: Converting Concurrent DRAM Indexes to Persistent-Memory Indexes** (Extended abstract of the SOSP 2019 paper), The 11th Annual Non-Volatile Memories Workshop (NVMW 2020).

[5] Rohan Kadekodi, **Se Kwon Lee**, Sanidhya Kashyap, Taesoo Kim, Aasheesh Kolli and Vijay Chidambaram, **SplitFS: Reducing Software Overhead in File Systems for Persistent Memory** (Extended abstract of the SOSP 2019 paper), The 11th Annual Non-Volatile Memories Workshop (NVMW 2020, **Memorable Paper Award**).

[6] Qingrui Liu, Joseph Izraelevitz, **Se Kwon Lee**, Michael L. Scott, Sam H. Noh, and Changhee Jung, **iDO: Compiler-Directed Failure Atomicity for Nonvolatile Memory** (Extended abstract of the MICRO 2018 paper), The 10th Annual Non-Volatile Memories Workshop (NVMW 2019).

[7] **Se Kwon Lee**, K. Hyun Lim, Hyunsub Song, Beomseok Nam, and Sam H. Noh, **WORT: Write Optimal Radix Tree for Persistent Memory Storage Systems** (Extended abstract of the FAST 2017 paper), The 8th Annual Non-Volatile Memories Workshop (NVMW 2017).

[8] Hyunsub Song, Young Je Moon, **Se Kwon Lee**, and Sam H. Noh, **Transforming Legacy File Systems into Persistent Memory Exploiting File Systems with MeLo@V**, The 8th Annual Non-Volatile Memories Workshop (NVMW 2017).

Posters

[1] Taeklim Kim, **Sekwon Lee**, Sergey Serebryakov, Harumi Kuno, Sharad Singhal, and Christopher J. Rossbach, **Improving GPU Utilization with a Zero-Copy Object Store for ML Applications**, Poster at the 30th ACM Symposium on Operating Systems Principles (SOSP 2024).

[2] Haris Volos, Kimberly Keeton, Yupu Zhang, Milind Chabbi, **Se Kwon Lee**, Mark Lillibridge, Yuvraj Patel, and Wei Zhang, **Memory-Oriented Distributed Computing at Rack Scale**, Poster at the 9th ACM Symposium on Cloud Computing (SOCC 2018).

[3] Rohan Kadekodi, **Se Kwon Lee**, Aasheesh Kolli, and Vijay Chidambaram, **Ledger: Increasing Performance of POSIX Applications on Persistent Memory**, Poster at the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2018).

[4] Haris Volos, Kimberly Keeton, Yupu Zhang, Milind Chabbi, **Se Kwon Lee**, Mark Lillibridge, Yuvraj Patel, and Wei Zhang, **Software challenges for persistent fabric-attached memory**, Poster at the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2018).

[5] Hyunsub Song, Young Je Moon, **Se Kwon Lee**, and Sam H. Noh, **Adapting Legacy File Systems to Work Efficiently for Persistent Memory based Storage**, Poster at the 14th USENIX Conference on File and Storage Technology (FAST 2016).

Patents

[1] Sam H. Noh, Young Je Moon, Hyunsub Song, and **Se Kwon Lee**, **Computing System and Method for Data Consistency**, Registration No. 10-1789933 (KO), Registration Date 10.18.2017.

Honors & Awards

2022	**UT Austin Graduate Dean’s Prestigious Fellowship Supplement**	*2022*
2021	**UT Austin Graduate Dean’s Prestigious Fellowship Supplement**	*2021*
2021	**Microsoft Research PhD Fellowship**	*2021-2023*
2020	**NVMW Memorable Paper Award**	*2020*

Skills

Programming Languages C, C++, Python, C#

System Programming Userspace system development, RDMA programming, Linux kernel

Professional Activities

- Program Committee for USENIX FAST (2028), ACM SoCC (2024, 2025, 2026), IEEE CLOUD (2025, 2026), ACM SYSTOR (2025)
- Reviewer for IEEE Transactions on Cloud Computing (2025)
- Reviewer for IEEE Transactions on Knowledge and Data Engineering, ACM Transactions on Architecture and Code Optimization, ACM Transactions on Storage (2024)
- Reviewer for IEEE Transactions on Computers (2023)
- Invited talk at IBM Research (May. 2023). Data-Intensive Systems for Emerging Memory and Disaggregation Technologies
- Volunteered as Slack Co-Chair for SOSP 2021
- Invited talk at Intel Labs (Oct. 2020). RECIPE : Converting Concurrent DRAM Indexes to Persistent-Memory Indexes

Reference

Available upon request